

ASYMPTOTICS OF THE DEFORMED COEFFICIENTS $h_m^{(c)}$

ABSTRACT. We give a self-contained analytic derivation of the large- m behavior of the coefficients $h_m^{(c)}$ defined by the deformed generating function $H^{(c)}(t) = e^{ct}/(1-t^2)$. The only input is the closed form on the unit disc; no individual coefficient is computed and no numerical evidence is invoked. We obtain an exact formula for $h_m^{(c)}$, an explicit bound on the error in the limiting expression, and the parity-split limits $\cosh c$ and $\sinh c$.

1. SETUP AND NOTATION

Fix $c \in \mathbb{C}$. We take as our single starting point the identity, valid as an equality of analytic functions on the open unit disc $\{t \in \mathbb{C} : |t| < 1\}$,

$$(1) \quad H^{(c)}(t) = \frac{e^{ct}}{1-t^2} = \sum_{m=0}^{\infty} h_m^{(c)} t^m.$$

This is what equation (D) produces for the constant sequence after deformation: solving $\frac{d}{dt} \log H^{(c)}(t) = c + \frac{2t}{1-t^2}$ with $H^{(c)}(0) = 1$ yields $\log H^{(c)}(t) = ct - \log(1-t^2)$ on a neighborhood of $t = 0$, hence (1), and the two sides, being analytic on the unit disc, agree throughout it. The numbers $h_m^{(c)}$ are exactly the Taylor coefficients in (1); their large- m behavior is the object of study. Nothing below uses the value of any particular $h_m^{(c)}$.

For $z \in \mathbb{C}$ and $m \geq 0$ write

$$E_m(z) = \sum_{k=0}^m \frac{z^k}{k!}, \quad R_m(z) = e^z - E_m(z) = \sum_{k=m+1}^{\infty} \frac{z^k}{k!},$$

so that $E_m(z)$ is the m -th Taylor polynomial of $\exp$ at 0 and $R_m(z)$ is its remainder.

2. AN EXACT FORMULA FOR THE COEFFICIENTS

Lemma 2.1. *For every $c \in \mathbb{C}$ and every $m \geq 0$,*

$$h_m^{(c)} = \frac{1}{2} E_m(c) + \frac{(-1)^m}{2} E_m(-c).$$

Proof. The partial-fraction identity

$$\frac{1}{1-t^2} = \frac{1}{2} \left(\frac{1}{1-t} + \frac{1}{1+t} \right),$$

an equality of rational functions on $\mathbb{C} \setminus \{\pm 1\}$, gives on the disc $|t| < 1$

$$H^{(c)}(t) = \frac{1}{2} \frac{e^{ct}}{1-t} + \frac{1}{2} \frac{e^{ct}}{1+t}.$$

For $|t| < 1$ the exponential series $e^{ct} = \sum_{k \geq 0} \frac{c^k}{k!} t^k$ converges absolutely (it is entire), and the geometric series $\frac{1}{1 \mp t} = \sum_{i \geq 0} (\pm 1)^i t^i$ converges absolutely. The product of two absolutely convergent power series is represented on the common disc of convergence by their Cauchy product, which again converges absolutely; hence the coefficient of t^m may be read off as a finite convolution. Thus

$$\begin{aligned} [t^m] \frac{e^{ct}}{1-t} &= \sum_{k=0}^m \frac{c^k}{k!} = E_m(c), \\ [t^m] \frac{e^{ct}}{1+t} &= \sum_{k=0}^m \frac{c^k}{k!} (-1)^{m-k} = (-1)^m \sum_{k=0}^m \frac{(-c)^k}{k!} = (-1)^m E_m(-c). \end{aligned}$$

Since the Taylor coefficients of the analytic function $H^{(c)}$ are unique, comparison with (1) yields the stated identity. $\square$

3. A TAIL ESTIMATE FOR THE EXPONENTIAL

Lemma 3.1. *For every $z \in \mathbb{C}$ and every $m \geq 0$,*

$$|R_m(z)| = |e^z - E_m(z)| \leq \frac{|z|^{m+1}}{(m+1)!} e^{|z|}.$$

Proof. Estimating the tail term by term,

$$|R_m(z)| \leq \sum_{k=m+1}^{\infty} \frac{|z|^k}{k!} = \frac{|z|^{m+1}}{(m+1)!} \sum_{j=0}^{\infty} \frac{(m+1)!}{(m+1+j)!} |z|^j.$$

For each $j \geq 0$,

$$\frac{(m+1)! j!}{(m+1+j)!} = \binom{m+1+j}{j}^{-1} \leq 1, \quad \text{so} \quad \frac{(m+1)!}{(m+1+j)!} \leq \frac{1}{j!}.$$

Therefore

$$|R_m(z)| \leq \frac{|z|^{m+1}}{(m+1)!} \sum_{j=0}^{\infty} \frac{|z|^j}{j!} = \frac{|z|^{m+1}}{(m+1)!} e^{|z|}. \quad \square$$

4. THE ASYMPTOTICS

Theorem 4.1. *For every $c \in \mathbb{C}$ and every $m \geq 0$,*

$$h_m^{(c)} = \frac{e^c + (-1)^m e^{-c}}{2} + \varepsilon_m(c), \quad |\varepsilon_m(c)| \leq \frac{|c|^{m+1}}{(m+1)!} e^{|c|}.$$

Consequently $\varepsilon_m(c) \rightarrow 0$ as $m \rightarrow \infty$, faster than any geometric sequence, and

$$\lim_{\substack{m \rightarrow \infty \\ m \text{ even}}} h_m^{(c)} = \frac{e^c + e^{-c}}{2} = \cosh c, \quad \lim_{\substack{m \rightarrow \infty \\ m \text{ odd}}} h_m^{(c)} = \frac{e^c - e^{-c}}{2} = \sinh c.$$

Proof. Substitute $E_m(\pm c) = e^{\pm c} - R_m(\pm c)$ into Lemma 2.1:

$$\begin{aligned} h_m^{(c)} &= \frac{1}{2}(e^c - R_m(c)) + \frac{(-1)^m}{2}(e^{-c} - R_m(-c)) \\ &= \frac{e^c + (-1)^m e^{-c}}{2} - \frac{1}{2}(R_m(c) + (-1)^m R_m(-c)). \end{aligned}$$

Define $\varepsilon_m(c) = -\frac{1}{2}(R_m(c) + (-1)^m R_m(-c))$. Because $|-c| = |c|$, Lemma 3.1 bounds each of $|R_m(c)|$ and $|R_m(-c)|$ by $\frac{|c|^{m+1}}{(m+1)!} e^{|c|}$, so

$$|\varepsilon_m(c)| \leq \frac{1}{2}(|R_m(c)| + |R_m(-c)|) \leq \frac{|c|^{m+1}}{(m+1)!} e^{|c|}.$$

The quantity $|c|^{m+1}/(m+1)!$ is the general term of the convergent series for $e^{|c|}$, so it tends to 0; moreover

$$\frac{|c|^{m+2}/(m+2)!}{|c|^{m+1}/(m+1)!} = \frac{|c|}{m+2} \xrightarrow{m \rightarrow \infty} 0,$$

which is the meaning of “faster than any geometric sequence.” Fixing the parity of m makes $(-1)^m$ constant, and letting $m \rightarrow \infty$ through that parity class sends $\varepsilon_m(c) \rightarrow 0$, giving the two limits. $\square$

Corollary 4.2 (the case $c = 1$). *For $c = 1$,*

$$h_m^{(1)} = \frac{e + (-1)^m e^{-1}}{2} + \varepsilon_m, \quad |\varepsilon_m| \leq \frac{e}{(m+1)!},$$

so $h_m^{(1)} \rightarrow \cosh 1 = \frac{e}{2} + \frac{1}{2e}$ over even m and $h_m^{(1)} \rightarrow \sinh 1 = \frac{e}{2} - \frac{1}{2e}$ over odd m .

Proof. Set $c = 1$ in Theorem 4.1; then $|c| = 1$ gives the bound $\frac{1^{m+1}}{(m+1)!} e^1 = \frac{e}{(m+1)!}$, and $\frac{1}{2}(e \pm e^{-1}) = \cosh 1$ or $\sinh 1$ according to parity. $\square$

5. REMARKS

Remark 1 (which pole supplies what). The decomposition in Lemma 2.1 separates the two simple poles of $H^{(c)}$ at $t = \pm 1$. The pole at $t = +1$ contributes the parity-independent term $e^c/2$ (the limit of $E_m(c)$); the pole at $t = -1$ contributes the alternating term $(-1)^m e^{-c}/2$. The alternation $(-1)^m$, and hence the splitting of the limit into the two distinct values $\cosh c$ and $\sinh c$, is produced solely by the *location* $t = -1$ of the second pole and is independent of c . The parameter c enters only through the numerator e^{ct} and therefore fixes nothing but the two residue magnitudes $e^{\pm c}/2$.

Remark 2 (independence from numerics). Every assertion follows from (1) together with the elementary Lemmas 2.1 and 3.1. In particular the limit is not an extrapolation from computed coefficients: for each finite m the deviation of $h_m^{(c)}$ from $\frac{1}{2}(e^c + (-1)^m e^{-c})$ is bounded explicitly by $\frac{|c|^{m+1}}{(m+1)!} e^{|c|}$.

Remark 3 (consistency with the small- t derivation). The identity (1) is established on a neighborhood of $t = 0$, and the asymptotics are extracted without ever evaluating $H^{(c)}$ at or near the poles. The geometric expansions $\frac{1}{1 \mp t} = \sum_{i \geq 0} (\pm 1)^i t^i$ used in Lemma 2.1 converge on exactly the disc $|t| < 1$ on which (1) holds. Thus the whole argument takes place inside the region where the derivation of (1) is valid; the role of the poles at $t = \pm 1$ is only to mark the boundary $|t| = 1$ of that disc, which is where the coefficient asymptotics are governed.